(*
Interlude: abstracting from the particular language
- to a theory of all abstract languages, and
- (all) possible interpreters (evaluators)

*)
(*
Signatures and Signed/Ranked Algebras *)
(*
Let us present a general framework for discussing Abstract Syntax

For simplicity, we confine our interest to a homogeneous algebraic framework: all expressions are of one *sort*. (The generalisation is called multi-sorted algebras.)
*)

(*
We first fix the symbols which any candidate (abstract) language will use:
Definition: Signature Σ - a set of symbols,
and for each symbol, its arity ≥ 0
*)

(* **Example**: Boolean expressions
`T` , `F` arity 0
¬ arity 1
∨, ∧ arity 2
*)

(* **Example**: Integer expressions
(a denumerable set of) numerals arity 0
`+` , `*` arity 2
unary minus `-` arity 1
*)

(*
We now discuss the meaning of expressions in any language whose symbols have been specified using a signature Σ.

Semantics

Definition (Σ-algebra)
A — a carrier set, and
for each 0-ary symbol in Σ, associate some element of A
for each k-ary symbol in Σ, associate a total function in $A^k \to A$

Notes:
- Not all elements of A require having a 0-ary symbol in Σ associated with them. That is, the carrier set may contain values (perhaps infinitely many) for which there is no syntax. Irrational real numbers are a good examples — there may be

no way of writing a symbol for each irrational number. In fact, we may not even be able to write an expression for each irrational number.
- Different 0-ary symbols in $\sum$ *do not necessarily* have to be associated with *different* elements — two *different* symbols can be mapped to the same value. Similarly for k-ary symbols, two *different* symbols can be mapped to the same function.
- Two $\sum$-algebras can have the same carrier set A, but differ on the interpretation of the symbols in the signature $\sum$. These would be considered *different* $\sum$-algebras.

*)

(*

Note: Symbols do not *inherently* carry meaning. We can assign whatever meanings we choose to .... but once we assign meaning by choosing a particular $\sum$-algebra, we have fixed the meanings of the symbols.

In particular, we can give
Standard examples of $\sum$-algebras, where the symbols are given meanings that we all expect and agree upon, e.g., binary symbol + means addition on naturals, the 0-ary symbol numeral 0 means the natural number zero.

But we can also give non-standard examples of $\sum$-algebras, where the meanings of symbols differ from what we normally expect, e.g., 0-ary symbol numeral 0 can be given the meaning "forty-two".
*)

(*
Notation:
If $\mathscr{A} = \langle A, \ldots \rangle$ is a $\sum$-algebra, and $a \in A$ is associated with 0-ary symbol c in $\sum$, we write $c_{\mathscr{A}}$ to denote the element $a \in A$, highlighting that this is the meaning associated with symbol c in algebra $\mathscr{A}$.
Similarly, if f is a k-ary symbol in $\sum$, and in $\sum$-algebra $\mathscr{A} = \langle A, \ldots \rangle$, f is associated with a total function $g : A^k \to A$, we write $f_{\mathscr{A}}$ to denote this function g.
*)

(* **Abstract Syntax, abstractly and formally**

Definition: (Carrier set $Tree_\Sigma$)
Suppose $\sum$ is a given signature.
The set $Tree_\Sigma$ is inductively defined as follows:

Base cases: for each 0-ary symbol c in $\sum$, a labelled node $\bullet c$ is in $Tree_\Sigma$
Induction cases: for *each* $k > 0$,
for each k-ary symbol f in $\sum$,
for *any* trees $t_1, \ldots, t_k$ (already) in $Tree_\Sigma$,
the tree consisting of
a labelled node $\bullet f$ at the root
with the k trees $t_1, \ldots, t_k$ ordered as subtrees below the root node
is in $Tree_\Sigma$.

Notes: The roots of $t_1, \ldots, t_k$ are the children, ordered 1...k, of the new root node $\bullet f$
The t_i need not be distinct.

The Abstract Syntax Tree Algebra: *Syntax as Semantics*
Definition: $\mathbf{Tree}_\Sigma$ is the Σ-algebra with carrier set $Tree_\Sigma$, where
- every 0-ary symbol c in Σ is interpreted as the labelled node $\bullet c$, which is in $Tree_\Sigma$, and
- every k-ary symbol f in Σ is interpreted as the *tree-forming function* that takes k trees $t_1, \ldots, t_k$ and creates the tree

```
   •f
  /  \
 t1...tk
```

in $Tree_\Sigma$ by sticking $t_1, \ldots, t_k$ as the subtrees below a new root node labelled $\bullet f$.
*)
(*
Structure-preserving functions between Σ-algebras
Σ-homomorphisms are *structure-preserving* functions *between* Σ-algebras which respect the symbol association (imposed with respect to the symbols in the signature Σ) by the semantics of the source and target Σ-algebras.

Definition (Σ-homomorphisms) Suppose Σ is a given signature.
Suppose $\mathscr{A} = \langle A, \ldots \rangle$ and $\mathscr{B} = \langle B, \ldots \rangle$ are both Σ-algebras (for the same signature Σ).
A total function $h : A \to B$ between the carrier sets of these algebras is called a Σ-homomorphism if
- *for all* 0-ary symbols c in Σ: $h(c_{\mathscr{A}}) = c_{\mathscr{B}}$
- *for all* k-ary (k >0) symbols f in Σ,
for all $a_1, \ldots a_k \in A$,
$$h(\ f_{\mathscr{A}}(a_1, \ldots, a_k)\ ) \ = \ f_{\mathscr{B}}(\ h(a_1), \ldots, h(a_k)\ )$$
*)
(*
Initiality Theorem
For any Σ-algebra, $\mathscr{B} = \langle B, \ldots \rangle$, there is a unique Σ-homomorphism
$i_{\mathscr{B}} : Tree_\Sigma \to B$
from the Σ-algebra $\mathbf{Tree}_\Sigma$ to the Σ-algebra $\mathscr{B}$.
Proof — by construction, define $i_{\mathscr{B}}$ to be a Σ-homomorphism.
Define $i_{\mathscr{B}} : Tree_\Sigma \to B$ as follows:
- for each 0-ary symbol c in Σ: $i_{\mathscr{B}}(\ \bullet c\ ) = c_{\mathscr{B}}$
- for each k-ary (k >0) symbol f in Σ,

```
   iB ( •f )  =  fB( iB(t1), ..., iB(tk) )
       /  \
      t1...tk
```

Uniqueness of $i_{\mathscr{B}}$

Suppose $j : Tree_\Sigma \to B$ is also a Σ-homomorphism from $\mathbf{Tree}_\Sigma$ to $\mathscr{B}$.

Proof by induction on (`ht` t) that *for all* t in $Tree_\Sigma$, $i_\mathscr{B}(\, t\,) = j(\, t\,)$

Base cases (`ht` t) = 0.

t must be of the form $\bullet c$ for some 0-ary symbol in Σ.

But for each 0-ary symbol c in Σ: $i_\mathscr{B}(\,\bullet c\,) = c_\mathscr{B}$ // defn of $i_\mathscr{B}$

$= j(\,\bullet c\,)$ // j is a Σ-homomorphism

// from $\mathbf{Tree}_\Sigma$ to $\mathscr{B}$

Induction Hypothesis:

Assume that for all t' in $Tree_\Sigma$ such that (`ht` t') $\leq n$,

$i_\mathscr{B}(\, t'\,) = j(\, t'\,)$

Induction Step: Consider t in $Tree_\Sigma$ such that (`ht` t) = n+1

Therefore t is of the form

```
  •f
 /  \
t1...tk
```

with (`ht` t_i) $\leq n$ $(1 \leq i \leq k)$

Now, by definition of $i_\mathscr{B}$,

for each k-ary symbol (k >0) f in Σ,

```
iB(  •f  )
    /  \
   t1...tk
```

$= f_\mathscr{B}(\, i_\mathscr{B}(\, t_1\,),\ldots, i_\mathscr{B}(\, t_k\,)\,)$ // definition of $i_\mathscr{B}$

$= f_\mathscr{B}(\, j(\, t_1\,),\ldots, j(\, t_k\,)\,)$. // by IH on each of $t_1\ldots t_k$

```
j(  •f  )     // j is a ∑-homomorphism from Tree∑ to B
   /  \
  t1...tk
```

*)